



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO SHIVA PETROL BUNK

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

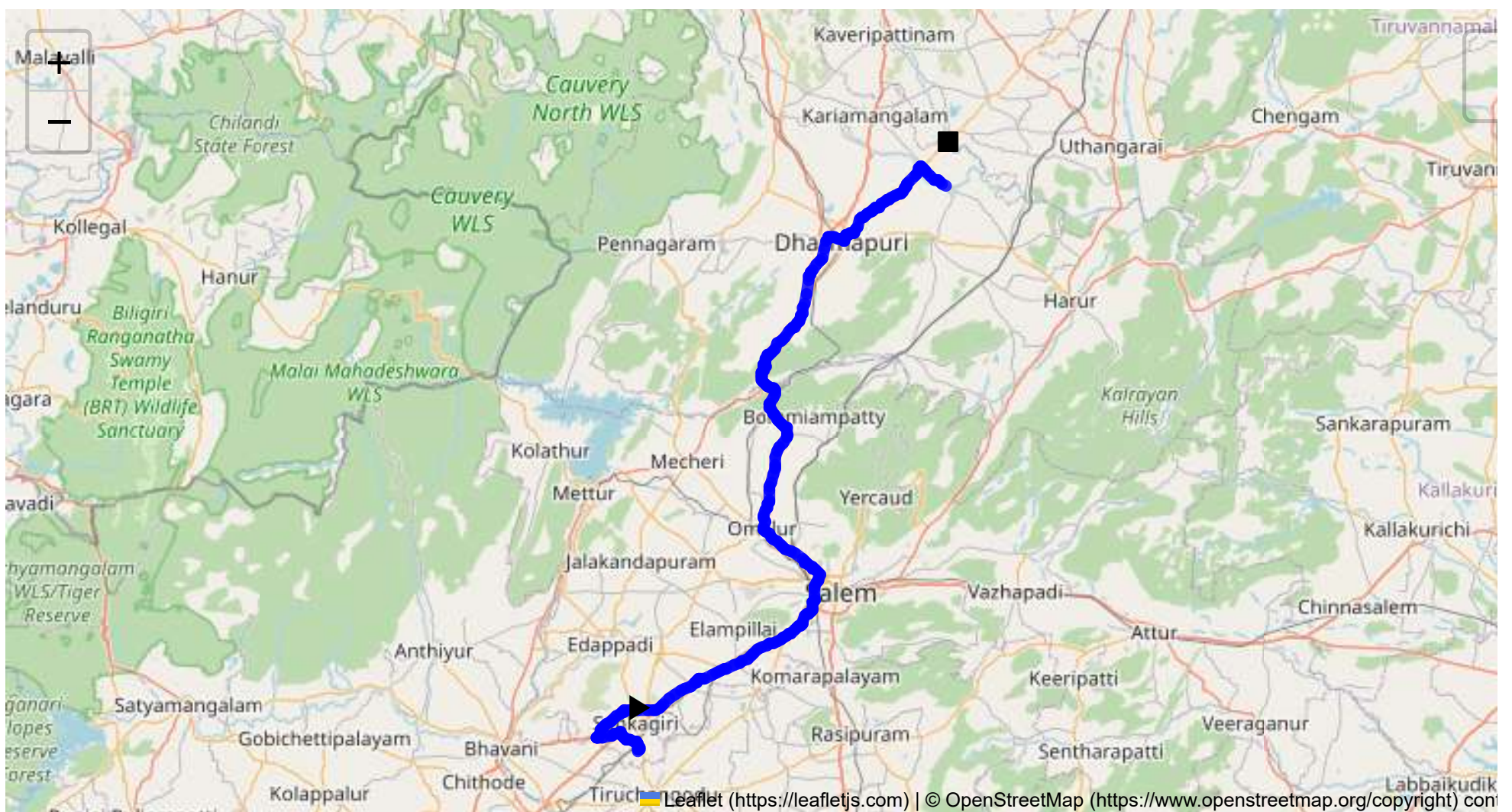
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 140.53 km
Estimated Duration: 2.7 hours
Adjusted Duration (Heavy Vehicle): 3.4 hours
Start: (11.4381, 77.8734)
End: (12.2114254, 78.305431)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map: The route from Sangagiri to Kambainallur via several intermediate points covers approximately 140.53 kilometers. It traverses through diverse terrains, including rural roads, town intersections, and national highways. Key towns along the route include Salem, Thoppur, and Dharmapuri, featuring a combination of state highways and local roads.

2. Typical Weather Conditions and Potential Weather-Related Hazards: The region generally experiences a tropical climate. Summers (March to June) can be hot, while the monsoon season (June to September) brings heavy rainfall, which might cause waterlogging and reduced visibility. Winter (November to February) is mild and generally more favorable for travel. Potential weather-related hazards include:

- Waterlogged roads during monsoons
- Reduced road grip due to rain
- Occasional fog in the morning during winters

3. Analysis of Traffic Patterns:

- **Peak Hours:** Morning (8-10 AM) and evening (5-7 PM) are typical peak hours.
- **Congestion-Prone Areas:** Salem is a major city on this route and often experiences traffic congestion, especially around the Bangalore bypass. Dharmapuri also sees elevated traffic levels during peak times.

4. Assessment of Road Quality and Infrastructure:

- **Sangagiri to Salem:** Roads are generally well-maintained but may have sections with pot-holes, especially post-monsoon.
- **Salem to Thoppur:** This segment includes national highways with good road quality, but watch for heavy traffic.
- **Thoppur to Kambainallur:** Combination of state highways and local roads, where the quality may vary; narrower roads as you approach Kambainallur.

5. Suggestions for Alternative Routes for Emergencies:

- In case of major congestion or roadblocks in Salem, consider using local roads bypassing busy segments.
- For emergencies, NH44 can help divert through alternative towns, although it adds distance.

6. Summary of Local Regulations Affecting Hazardous Material Transport:

- Permits are required for transporting hazardous materials.
- Restrictions on the transport of such materials through densely populated areas during peak hours.
- Compliance with safety standards for vehicle maintenance and driver training is mandatory.

7. Overview of Historical Incidents: While specific historical incident data is limited, common incidents involve traffic collisions in high-density areas like Salem and Thoppur, often due to congestion and driver fatigue.

8. Environmental Considerations and Sensitive Areas:

- The route passes near agricultural lands which need protection from spillage or pollution.
- Dense population areas in Salem and Dharmapuri demand extra caution to prevent contamination risks.

9. Analysis of Communication Coverage: Generally good coverage along the route, but potential dead zones might exist in rural stretches beyond Thoppur and near Kambainallur. Wireless Internet and mobile

connections may drop intermittently.

10. Estimated Emergency Response Times:

- Urban segments (e.g., Salem): Approximately 15-30 minutes for emergency services due to proximity to facilities.
- Rural segments: 30-60 minutes owing to longer travel distances for first responders.

12. Overall Summary of Risk Assessment: The route entails medium risk due to varied road conditions, significant urban traffic congestion, and monsoon-related weather hazards. Continuous vigilance and advanced route planning are essential for minimizing risks, especially when transporting hazardous materials. Adequate preparation includes real-time navigation tools to adapt to traffic and weather conditions, adherence to safety regulations, and ensuring communication is constant and effective.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	12.13658, 78.16171	15 KM/Hr
1	Roundabout	High	12.14589, 78.17156	15 KM/Hr
2	Roundabout	High	12.23740, 78.27029	15 KM/Hr
3	Turn	High	11.43968, 77.87345	15 KM/Hr
4	Turn	High	11.44029, 77.87544	15 KM/Hr
5	Turn	High	11.44898, 77.87410	15 KM/Hr
6	Turn	High	11.45352, 77.85698	15 KM/Hr
7	Turn	Medium	11.45764, 77.85219	30 KM/Hr
8	Turn	High	11.46303, 77.84945	15 KM/Hr
9	Turn	Medium	11.45543, 77.81740	30 KM/Hr
10	Turn	High	11.45631, 77.81668	15 KM/Hr
11	Turn	Medium	11.53573, 77.95794	30 KM/Hr
12	Turn	Medium	11.63056, 78.11726	30 KM/Hr
13	Turn	Medium	11.64183, 78.11810	30 KM/Hr
14	Turn	Medium	11.64338, 78.11998	30 KM/Hr
15	Turn	Medium	11.68116, 78.12695	30 KM/Hr
16	Turn	Medium	11.70148, 78.10372	30 KM/Hr
17	Turn	Medium	11.74024, 78.05065	30 KM/Hr
18	Turn	Medium	11.77961, 78.05775	30 KM/Hr
19	Turn	Medium	11.86873, 78.08374	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
20	Turn	Medium	11.87966, 78.08187	30 KM/Hr
21	Turn	Medium	11.92609, 78.06513	30 KM/Hr
22	Turn	Medium	11.93710, 78.05390	30 KM/Hr
23	Turn	High	11.95824, 78.04981	15 KM/Hr
24	Turn	Medium	11.95992, 78.04947	30 KM/Hr
25	Turn	Medium	11.96320, 78.05139	30 KM/Hr
26	Turn	High	11.96401, 78.05134	15 KM/Hr
27	Turn	Medium	11.96504, 78.04975	30 KM/Hr
28	Turn	High	11.96597, 78.04920	15 KM/Hr
29	Turn	Medium	11.96679, 78.04942	30 KM/Hr
30	Turn	Medium	11.96804, 78.05080	30 KM/Hr
31	Turn	Medium	12.14026, 78.13955	30 KM/Hr
32	Turn	Medium	12.14048, 78.13998	30 KM/Hr
33	Turn	High	12.14485, 78.16498	15 KM/Hr
34	Turn	Medium	12.14695, 78.17515	30 KM/Hr
35	Turn	Medium	12.20752, 78.24730	30 KM/Hr
36	Turn	Medium	12.23366, 78.27442	30 KM/Hr
37	Turn	Medium	12.21808, 78.29244	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
2	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
3	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium
4	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium
5	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium
8	hospital	Vinayaka Mission's Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
11	hospital	Bharathi Hospital, Salem	11.6492647, 78.1204131	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
12	hospital	Sisu Hospital Inida Ltd	11.6505276, 78.1205003	30 km/h	Medium
13	hospital	Aishwarya Hospital	11.6716291, 78.1267807	30 km/h	Medium
14	hospital	C.M. Hospital	11.6747345, 78.1255034	30 km/h	Medium
16	hospital	Sri Gokulam Speciality Hospital	11.6788724, 78.1274258	30 km/h	Medium
17	pharmacy	Healthy Pharmacy	11.6762714, 78.127744	30 km/h	Medium
18	pharmacy	Thulasi Pharmacy	11.6761991, 78.1277377	30 km/h	Medium
19	hospital	Global Medical Center	11.6867049, 78.1186697	30 km/h	Medium
20	hospital	Manipal Hospital	11.703072, 78.1026612	30 km/h	Medium
23	pharmacy	MJ Medicals	11.7232728, 78.0698965	30 km/h	Medium
24	pharmacy	Salem Medicals	11.7343192, 78.0566844	30 km/h	Medium
26	hospital	Vinayaka Child Hospital	11.745812, 78.0480225	30 km/h	Medium
27	hospital	Children Hospital	11.7458377, 78.0480205	30 km/h	Medium
28	clinic	Manimuthu Skin Clinic	11.7521189, 78.0484058	30 km/h	Medium
29	hospital	St. Joseph's Speciality Hospital	11.761611, 78.0522344	30 km/h	Medium
31	hospital	Anugraha Multi Speciality Hospital	11.773549, 78.0546408	30 km/h	Medium
33	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium
34	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium
36	hospital	Vijaya Hospital	12.1399758, 78.1532115	30 km/h	Medium
37	clinic	Bee Tee Clinic	12.138889, 78.1576748	30 km/h	Medium
38	clinic	Sheela Clinic	12.138725, 78.1573436	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
39	clinic	Suguna Clinic	12.1386752, 78.1575468	30 km/h	Medium
40	hospital	Navina Hospital	12.1367521, 78.1610296	30 km/h	Medium
41	clinic	Bharani Neurologist	12.135969, 78.16255	30 km/h	Medium
42	hospital	D.N.V Hospital	12.1351328, 78.162869	30 km/h	Medium
43	hospital	Pragadeesh Hospital	12.135284, 78.1591867	30 km/h	Medium
44	clinic	Ragavendra Physiotherapy Clinic	12.135569, 78.1635247	30 km/h	Medium
45	hospital	Kamatchi Hospital	12.1350608, 78.1594841	30 km/h	Medium
46	clinic	Malar Siddha Clinic	12.1350485, 78.159226	30 km/h	Medium
47	hospital	Kamala Nehru Hospital	12.13404, 78.1633476	30 km/h	Medium
48	clinic	Govindaraj Clinic	12.1344914, 78.1642372	30 km/h	Medium
49	hospital	Sri Priya Hospital	12.135347, 78.164391	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
6	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium
7	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium
9	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium
10	school	DIET - Salem	11.6144158, 78.1025078	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
15	college	Sona College	11.6768, 78.1257052	30 km/h	Medium
21	college	Govt Engineering College	11.7128537, 78.0877084	30 km/h	Medium
22	university	Periyar University	11.7186758, 78.0771089	30 km/h	Medium
25	school	Fathima Hr Sec School	11.7410508, 78.0479272	30 km/h	Medium
30	school	John Britto Matriculation Higher Secondary School	11.7676907, 78.0513092	30 km/h	Medium
32	college	Dhirajlal Gandhi College of Technology	11.7968014, 78.0562095	30 km/h	Medium
35	college	Laxminarayana Polytechnic College	12.0421277, 78.103203	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.13658, 78.16171



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.14589, 78.17156



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.23740, 78.27029



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44898, 77.87410



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45352, 77.85698



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45764, 77.85219



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.46303, 77.84945



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45543, 77.81740



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45631, 77.81668



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.53573, 77.95794



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.63056, 78.11726



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64338, 78.11998



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.68116, 78.12695



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.70148, 78.10372



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.74024, 78.05065



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.77961, 78.05775



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Speed Limit: 30 KM/Hr
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Coordinates: 11.92609, 78.06513



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Speed Limit: 30 KM/Hr
Coordinates: 11.93710, 78.05390



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Speed Limit: 15 KM/Hr

Coordinates: 11.95824, 78.04981



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.95992, 78.04947



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96320, 78.05139



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96401, 78.05134



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96504, 78.04975



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.96597, 78.04920



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.96679, 78.04942



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.96804, 78.05080



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.14026, 78.13955



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.14048, 78.13998



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.14485, 78.16498



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.14695, 78.17515



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.20752, 78.24730



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Risk Level: Medium
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Risk Level: Medium
Speed Limit: 30 KM/Hr
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